Futures Contracts

Fundamentals of Trading

FINM 37000 - Futures And Related Derivatives

Eric Patterson, Fall 2025

Futures Contracts

Comparison with Forwards

Forwards

- Agreement to buy/sell at future time at agreed price
- OTC
- Custom contract specs
- Settled at expiration
  - Gains/losses deferred

Futures

- Agreement to buy/sell at future time at agreed price
- Exchange Traded
- Standardized contract specs
- Settled daily (mark-to-market)
  - Gains/losses assessed daily

Cashflows

Futures vs. Forwards

- Forward, buy or sell at time T for price K
  - Time of trade: no cashflow
  - Time T: Buyer pays seller K, seller delivers asset
- Future, buy or sell at time T for price K
  - Time of trade: no cashflow between traders, margin accounts must be funded
  - End of trade day: adjust margin accounts (mark to market), position value resets to 0
  - End of next day: mark to market, position value resets
  - …
  - Time T: mark to market and process delivery if physically settled

Mark to Market

Daily futures cash flows

- Buy CLV5 at noon on 2025-09-09.
- See example in Jupyter notebook. Key concepts:
    - Mark to market/margin
    - Settlement (vs. Close)
    - Open Interest
    - Price limits

Mechanics of a futures trade

Example: July 2025 Soybeans (ZSN5)

- 1 Futures contract traded at 1050’2 = 1050.25
  - Pay $0 now
  - Fund margin account
  - Buyer: If held through expiration, receive soybeans
  - Seller: If held through expiration, deliver soybeans
- Contract Specs
  - Settlement date: 14 July 2025
  - Contract unit: 5,000 bushels
  - Price unit: U.S. cents per bushel
  - Minimum Price Fluctuation: 1/4 cent per bushel

Mechanics of a futures trade

Example: July 2025 Soybeans (ZSN5)

- 1 Futures contract traded at 1050’2
- End of day: settlement price 1060’4 = 1060.50
    - Buyer/long: Margin +10.25¢/per bushel x 5000 bushels = $512.50
    - Seller/short: Margin -$512.5

Mechanics of a futures trade

Example: July 2025 Soybeans (ZSN5)

- Futures contract settled at 1060’4
- Next day
  - Buyer exits at a price 1055’4
    - Original Buyer sells
      - No position
      - Margin -5¢/per bushel x 5000 bushels = -$250
    - New buyer: long 1

Open Interest vs. Trade Volume

Example: July 2025 Soybeans (ZSN5)

- Day 1
    - Volume: +1
    - Open Interest: +1
- Day 2
    - Volume: +1
    - Open Interest: unchanged
    - Suppose seller sells another contract to the new buyer
        - Daily volume: +2
        - Open Interest: +1 (+2 including Day 1)

Delivery and the Basis

Physical Delivery

- Futures position held through delivery
  - Long: receive delivery of physical
  - Short: deliver physical (warehouse receipt or shipping certificate)

Delivery and the Basis

Physical Delivery

- July Soybean example: 5000 BU/contract
    - Trades through July 14th (final settlement date)
    - Position Day
        - Short gives exchange intent to deliver between June 27th and July 15th
            - Grades: Yellow #1 (+6¢/BU), Yellow #2, Yellow #3 (-6¢/BU)
            - Locations: 8 different locations +0-16.25¢/BU
        - Exchange selects long to take delivery (time priority)
    - Notification Day (Position+1): Notify long/short they have been matched
    - Delivery Day (Position+2): Long pays short through clearing, receives delivery

Delivery and the Basis

Physical Delivery

- Basis trade buys and sells physical versus futures for potential delivery
- Examples
  - Commodities: grades and locations for delivery
  - Treasuries: Treasury notes with remaining term to maturity meeting specified time range (e.g., 6.5-8 years for Ten year futures), with delivery price adjustment
  - FX: deliver currency

Delivery and the Basis

Financial Settlement

- No physical delivery.
- Observed price/index/rate determines financial value.
- Examples
    - SOFR
    - S&P 500 Index

Who Trades, Why, and How Much

- Who
  - hedgers
  - speculators
  - market makers
- Why: Take or reduce risks from
  - Commodities: Grains, Meats, Energy, Metal
  - Fixed income: Treasuries, SOFR, Fed Funds
  - Equity Indexes: S&P, Nasdaq, Dow, Russell, Nikkei
  - FX, Electricity

Who Trades, Why, and How Much

	Soybeans	Corn	Crude	E-mini S&P	Ten Years	3-month SOFR
Contract Size	5000 Bushels	5000 Bushels	1000 Barrels	$50 x S&P Index	$100,000 face value at maturity	$2500 x IMM index
Volume/day	100k-700k	200k-1mill	500k-2.5 mill	1-5 mill	1.5-6 mill	2-15 mill
Open Interest	800-900k	1.5-2 mill	1.5-2 mill	2-3 mill	4-5 mill	10-12 mill